ELEC5305 Acoustics, Speech and Signal Processing

Project Report

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1 Introduction

Understanding emotional expression has had a rapid rise in research importance after the rise in artificial intelligence applications. The automated emotion detection industry already holds a massive 48.71 billion USD value and is expected to rise to 141 billion USD in the next few years. While a large fraction of this is based on computer vision, a good portion of emotion-related applications only have access to acoustic data, hence the focus of our study. Some examples of acoustic emotion detection include tracking customer impressions of customer service from their voice, diagnosing conditions such as clinical depression through sessions with patients, and training LLM voice-interactive models to respond appropriately to users.

Until recently, researchers relied on a limited number of hand-crafted acoustic features for acoustic emotion detection, including F0 (pitch), speaker energy, spectral centroid and MFCCs. However, recent developments in deep learning have produced SSL (self-supervised learning) feature extractors, which can be trained on vast amounts of data autonomously to produce layers of features (low level to high level) that perform far better when these features are fed to machine learning models. Two such models used in this study are WavLM Base+ and UniSpeech-SAT Base+, both designed by Microsoft.

Prior work strongly supports self-supervised speech representations for emotion recognition, but it judges these models almost entirely by how accurately they predict the intended emotion label and, in some studies, by which layers predict it best. Human listeners are skilled at recognising emotion from the voice, but they are not perfect: agreement between listeners varies widely from one recording to another, and some expressions are genuinely ambiguous. It has not yet been examined whether SSL-based recognition struggles with the same recordings that listeners find ambiguous, or at which depth of the network any such correspondence is strongest. This study first replicates established methods (frozen SSL feature extraction, layer-wise probing and speaker-independent evaluation) and then extends them to address three questions:

RQ1: To what extent does the confidence of classifiers built on WavLM and UniSpeech-SAT features match human listeners' agreement on each recording?

RQ2: Which layers of these models hold the most emotion information, and which layers best match human perceptual difficulty?

RQ3: How do interpretable hand-crafted acoustic features compare with the SSL features on both of these measures?

The study uses the audio-only speech recordings from RAVDESS: 1,440 clips of 24 professional actors portraying eight emotions, each rated by about ten listeners. Both SSL models are used without further training, with a simple linear classifier on top, so that any differences reflect the features rather than the classifier. Because the recordings are acted, the study concerns the emotion the actor intended and the emotion listeners perceived, not what the speaker actually felt.

The results show that the middle layers of both SSL models hold the most emotion information, and that model confidence moderately matches how easily listeners recognised each recording, most closely in those same middle layers. The hand-crafted features reflect which *emotions* listeners find difficult, whereas the SSL features better reflect which individual *recordings* listeners find ambiguous. Section 2 reviews related work, Section 3 describes the methods, Section 4 presents the results, Section 5 discusses them, and Section 6 concludes.

2 Literature Review

The literature review is divided into five sections, all of which were important for this project. Section 2.1 focuses on the importance of different acoustic cues and how they relate to emotion, which was important for developing the hand-crafted baseline features for this study. Section 2.2 discusses the two SSL models used in this study, to show how powerful modern deep-learning feature extractors are at emotion detection.

Section 2.3 discusses which layers of features from these chosen SSL models have been found to be the best for emotion detection. Section 2.4 discusses our dataset, RAVDESS, and how it adds to the novelty of

our project by including human ambiguity in emotion detection. The final section of the literature review, Section 2.5, discusses why it is important to develop models that depend only on audio features instead of memorising speaker identity for emotion detection applications.

2.1 Acoustic cues to understand vocal emotions

Research on vocal emotion consistently shows that emotions differing in energy are much easier to tell apart than emotions of similar energy. High-energy emotions are marked by higher pitch, faster speech and greater loudness, whereas emotions with similar energy levels share many of these cues and are harder to separate [1], [2].

Scherer's review frames vocal emotion using the Brunswikian lens model, in which communication happens in three stages [1]. The speaker's emotional state is first *encoded* into measurable acoustic (distal) cues such as F0, loudness and voice quality; these cues are *transmitted* through speech; and the listener *decodes* them into a perceived emotion. Listeners do not perceive the distal values directly, but approximate proximal impressions, for example hearing a voice as "high-pitched" rather than as 200 Hz. Across about 30 studies, listeners recognised emotion from voice alone with roughly 60% accuracy, and 66% in a nine-country cross-cultural study [1]. Sadness and anger were easiest to recognise, fear intermediate, and joy and disgust hardest.

These cues mainly convey arousal. F0, energy and speech rate reliably signal how activated a speaker is, whereas valence (pleasant vs unpleasant) appears to depend more on voice quality and articulation, reflected in the spectrum and formants [1]. Juslin and Laukka's meta-analysis of 104 vocal-expression and 41 music-performance studies confirmed that listeners decode intended emotions well above chance (about 70% raw accuracy) and that each emotion has a typical cue profile (Table 1) [2]. Eyben et al. reached the same conclusion computationally: using leave-one-speaker-out evaluation, binary arousal was recognised with about 80% unweighted average recall, but binary valence with only 65–66% [3].

Table 1: Typical vocal cue patterns by emotion (adapted from [2]).

Emotion	Speech rate	Intensity	F0 level	F0 variability	High-freq. energy
Anger	Fast	High	High	Much	Much
Fear	Fast	Low (except panic)	High	Little	Little
Happiness	Fast	Medium–high	High	Much	Medium
Sadness	Slow	Low	Low	Little	Little
Tenderness	Slow	Low	Low	Little	Little

Rather than using thousands of brute-force features, Eyben et al. proposed GeMAPS, a compact set chosen because the features reflect physiological changes in voice production, performed well in earlier work and have theoretical meaning [3]. The set covers frequency, energy and spectral-balance parameters plus a few low-order MFCCs, which consistently improved recognition over prosody alone. The compact set performed close to sets with thousands of features, although F0 became less reliable in noise and valence remained difficult.

Acoustic cues also vary strongly between speakers. On RAVDESS itself, Major and Chatterjee found that emotion significantly affected every prosodic cue, that strong-intensity portrayals were higher in pitch, louder and longer, and that similar emotions (e.g. sad and disgust) differed on fewer cues than contrasting ones [4]. Accounting for the individual talker raised the variance explained in mean loudness from 51% to 85% and in mean F0 from 75% to 93% [4].

Together, these studies justify a compact, interpretable baseline of prosodic, spectral and cepstral features for our study, and predict that confusions will occur mainly between emotions of similar arousal. However, they rely on hand-designed features, mostly use acted portrayals, and predate representations learned automatically from raw audio, which are reviewed in Section 2.2.

2.2 Self-supervised speech models (SSL) for emotion recognition

Hand-crafted features are each defined by a formula for a known cue, such as pitch or energy. Self-supervised learning (SSL) models instead learn their own representation directly from the raw waveform, using large amounts of unlabelled speech. Features from pretrained SSL models such as WavLM and UniSpeech-SAT consistently outperform hand-crafted features for emotion recognition [5], [7]–[10].

WavLM [5] is a general-purpose speech model. A convolutional encoder turns the waveform into frames about every 20 ms, followed by a Transformer. The Base+ version has 12 Transformer layers with 768-dimensional outputs. It is trained with masked speech prediction from HuBERT [6], where the model predicts hidden frames from the surrounding context, together with denoising, where it must recover the main speaker's content from audio mixed with noise or a second speaker. The layer weights learned in [5] showed that speaker tasks rely mostly on the lower and middle layers, while content recognition relies on the top layers.

UniSpeech-SAT [7] has the same size but adds speaker-aware pretraining through an utterance-wise contrastive loss and utterance mixing. This gave large gains in speaker tasks and improved IEMOCAP emotion accuracy to 68.48%, compared with 64.92% for HuBERT Base. However, WavLM Base+ reached a nearly identical 68.65% [5], which suggests explicit speaker-aware training adds little for emotion.

SUPERB [8] standardised how such models are compared. The pretrained model is kept frozen, and a lightweight downstream model is trained on a learned weighted sum of all hidden layers. Because the weighted sum mixes every layer, it does not directly show which layer holds the emotion information. EMO-SUPERB [9] extended this into a dedicated emotion benchmark. It released fixed data partitions to make results reproducible, reported macro-F1 to handle class imbalance, and trained on soft labels built from annotator vote distributions.

The closest work to this project is Pratihast's thesis [10]. It compared hand-crafted features, OpenL3 and 17 frozen SSL models on MSP-Podcast, IEMOCAP and MSP-IMPROV. On MSP-Podcast, WavLM reached concordance correlation coefficients of 0.546, 0.264 and 0.433 for arousal, valence and dominance, compared with 0.213, 0.047 and 0.121 for MFCC features. WavLM and UniSpeech-SAT were consistently among the best models. Replacing mean pooling with a two-step sequence reduction improved valence, showing that averaging over time loses some timing detail. However, embeddings were taken only from the final layer, and predictions were not compared with human perceptual agreement.

2.3 Layer-wise analysis of SSL models

The layers of an SSL model hold different kinds of information. Pasad et al. [11] examined wav2vec 2.0 [12] layer by layer, using canonical correlation analysis, mutual information and simple probes. Each layer was compared with mel-spectrogram features, phone identity, word identity and word meaning. They found an acoustic-to-linguistic hierarchy. The last CNN layers and first Transformer layers resemble the mel spectrogram, deeper layers encode phones and then words, and the final layers of the pretrained model drift back toward the input. This autoencoder-like behaviour comes from the pretraining task, and fine-tuning for speech recognition removes it, so the final layers then encode words better. The study focused on linguistic content and did not test emotion or speaker information.

Pepino et al. [13] applied frozen wav2vec 2.0 features to emotion recognition on IEMOCAP and RAVDESS, with a downstream network that learned a weighted average of all layers. The weighted combination beat using only the CNN output or only the final layer, and the learned weights favoured the middle layers. The authors suggest these layers are already contextualised but not yet specialised to the pretraining task. Fine-tuning for speech recognition reduced emotion performance, showing that information irrelevant to words can still be important for emotion. Adding eGeMAPS features [3] gave further gains, so the SSL features did not capture all of the useful prosodic information. On RAVDESS, however, calm and neutral were merged into one class and only two actors were used for testing. Speaker normalisation also used the test speakers' own statistics.

Other studies point the same way. In the SUPERB weighted sums for WavLM [5], speaker tasks relied on lower layers and content tasks on upper layers. For emotion, EMO-SUPERB [9] found no clear shared pattern in the learned layer weights across models. This shows that learned weights are hard to interpret, and it is a

reason to probe each layer on its own. Work reviewed in [10] also reports that intermediate layers of WavLM and HuBERT outperform final layers for paralinguistic tasks.

Together, these findings suggest that emotion information should peak in the middle of the network rather than at the top. However, these studies measure only classification performance per layer. None asks which layer's confidence best follows how difficult human listeners found each clip.

2.4 Human perception, RAVDESS validation and ambiguity

Listeners recognise vocal emotion well above chance but far from perfectly. Scherer [1] reports an average accuracy of about 60%, roughly five times chance, with sadness and anger the easiest emotions to recognise in the voice and joy and disgust among the hardest. Juslin and Laukka [2] found raw accuracy of about 70% in five-choice tasks, but noted that lab studies use clear, acted portrayals, so real-life accuracy is likely to be lower. Listeners also disagree more on some clips than on others. Majority-vote labels hide this disagreement.

RAVDESS [14], the dataset used in this project, is unusual because it records this disagreement for every clip. It contains recordings of 24 professional actors (12 male, 12 female) with a neutral North American accent. The speech set covers eight emotions, each at normal and strong intensity except neutral. In validation, 247 raters made a forced choice of emotion category, giving about 10 ratings per stimulus, and also rated intensity and genuineness on 5-point scales. Each clip therefore has a proportion of raters who chose the intended emotion. Audio-only recognition was clearly harder than audio-visual. Mean audio-only accuracy was 0.60, and for strong speech, disgust and happiness fell from 88% and 84% audio-visual to 54% and 44% audio-only. Agreement among raters was only moderate, with Fleiss' kappa of 0.62 for strong and 0.53 for normal-intensity speech. The most common confusions were calm heard as happy (19% of responses), sad as neutral or calm (17%) and happy as neutral or calm (14%). An important scoring rule is that neutral and calm responses were accepted as correct for both neutral and calm expressions, because the authors did not expect listeners to separate them reliably. The authors also note several limitations: acted expressions may be exaggerated, raters were mostly undergraduate students, and forced choice can bias answers.

Recent SER work has begun to treat this ambiguity as information rather than noise. EMO-SUPERB [9] observed that annotators often disagree or describe mixed emotions, and it trained on soft labels built from vote distributions. Pratihast [10] noted that categorical labels struggle to represent blended or ambiguous states, which is one motivation for dimensional models. In both cases, however, human disagreement is used as a training target. Few studies use it the other way around, as a reference for testing whether a model's uncertainty matches human uncertainty clip by clip.

2.5 Speaker-independent evaluation

If the same speaker appears in both the training and test sets, an emotion classifier can partly learn the speaker's voice instead of the emotion, and its scores will be inflated. This risk is especially high for SSL features, which strongly encode speaker identity [5], [7]. Recent benchmarks therefore hold out whole speakers. SUPERB [8] evaluates emotion on IEMOCAP using leave-one-session-out cross-validation, where each session contains different speakers. EMO-SUPERB [9] found that inconsistent data partitions made many published results hard to compare, and it released fixed partitions for each dataset. Pratihast [10] also used speaker-independent cross-validation for IEMOCAP and MSP-IMPROV. Speaker effects remain large even when speakers are held out. Pepino et al. [13] found that normalising features with each test speaker's own statistics clearly improved RAVDESS results, but this uses information from the test speakers, and their split tested only two actors. Leave-one-actor-out cross-validation addresses both problems. Each actor is tested once, using a model trained on the other 23, so every actor contributes a test result and the variation across speakers can be reported. This project follows that design.

3 Methods

3.1 Data and human perceptual ratings

The RAVDESS dataset was used [14]. It consists of 1,440 clips from 24 professional actors with an equal male–female ratio. Each actor read two lines ("Kids are talking by the door", "Dogs are sitting by the door") twice for eight different emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised). All emotions (excluding neutral) are recorded at normal and strong intensity by each actor, so 60 clips can be attributed to each of the 24 actors. Python parsing was used to extract information such as file name, gender and actor identity number in `ravdess_meta.py`, for easier use when we perform feature extraction later.

From the RAVDESS validation data, we obtained the human agreement for each clip, which shows what emotions each of the 10 voters selected based on their perception of emotion from the audio clips [14]. Each clip then received a score for the intended emotion (PropCorr) showing what proportion of the voters chose the intended emotion. This data is important for our study's novelty, as it allows us to compare human emotion perception ambiguity with that of modern SSL models. Mean agreement was 0.62, close to the 0.60 reported by the authors. Two properties of this measure affect the analysis. First, neutral and calm responses were scored as correct for each other, so agreement for these two classes is lenient compared with the strict scoring of the classifiers; this is tested in a robustness check (Section 3.5). Second, with about 10 raters per clip the measure is noisy: two copies of the same actor 7 happy recording received 0.7 and 0.2. This limits how strongly any model can correlate with human agreement.

Python (pandas) scripts built a single metadata table (1,440 rows, checked for 60 clips per actor) and a fold table assigning each clip to its actor's fold. Both tables were shared by every system, so all systems saw exactly the same train and test splits.

At this point, the data preprocessing steps were complete.

3.2 Interpretable acoustic features (MATLAB)

In Section 2.1 we discussed explainable hand-crafted acoustic cues that were used in classical acoustic and emotion detection studies. We therefore first made our feature baseline, consisting of a few explainable distal features.

Taking inspiration from our literature review, we chose pitch, loudness, timing and spectral shape, as these were identified as most relevant to vocal emotion [1]–[3]. Each clip was converted to mono and resampled from 48 kHz to 16 kHz (the rate expected by the SSL models). Short-time analysis used a 25 ms Hamming window with a 10 ms hop, following the STFT framing of the Week 4 lab. Frames within 35 dB of the loudest frame were treated as speech, a simple energy-based voice activity detector based on the Week 6C lab, and all statistics were computed over speech frames only. Table 2 lists the 36 features.

F0 was estimated with MATLAB's `pitch()` function using the normalised correlation function (NCF) method, an autocorrelation-based estimator linked to the source-filter model of Week 7. A 40 ms window was used so that at least two periods of the lowest expected F0 fit in each frame. F0 variation and range were expressed in semitones relative to 100 Hz, so that male and female voices are compared on the same perceptual scale. MFCCs were computed with `mfcc()`, which applies a mel filterbank, log compression and DCT, extending the filterbank analysis of Week 6A.

The F0 features required several iterations. In the first version, `pitch()` returned a value for every frame, including unvoiced sounds such as /s/ and breath, and low male voices produced octave errors; the mean male F0 came out at about 207 Hz, far above the expected 100–150 Hz. Three fixes were added: a harmonic-ratio check (`harmonicRatio()`) so that only voiced frames are kept, a 5-point median filter to remove one-frame octave jumps, and semitone units for spread. A threshold of 0.8 then proved too strict, leaving 135 male clips with no voiced frames, while actor 9 still averaged 436 Hz because of octave errors. The final version lowered the voicing threshold to 0.65 and used gender-specific search ranges (60–350 Hz for male and 100–500 Hz for female actors, similar to Praat defaults), after which both genders fell within the expected ranges. Any remaining missing values were filled inside each training fold in Python, so no information from the test actor

Table 2: The 36 interpretable acoustic features and the course labs they build on.

Feature group	Count	Computation	Course lab
F0 (pitch)	3	mean (Hz); std and 5–95% range (semitones re 100 Hz); voiced frames only	Week 7A/B source-filter, autocorrelation F0
Intensity	3	frame RMS in dB: mean, std, 5–95% range over speech frames	Week 6C frame RMS
Duration	2	speech duration (s); speech-to-total frame ratio from energy VAD	Week 6C energy VAD
Spectral centroid	2	mean and std of per-frame centroid of the FFT magnitude	Week 4 STFT framing
MFCC 1–13	26	mean and std of each coefficient over speech frames	Week 6A filterbank analysis

was used.

3.3 Frozen self-supervised representations (Python)

We chose two of the best SSL models from Pratihast's PhD thesis, WavLM Base+ [5] and UniSpeech-SAT Base+ [7], designed by Microsoft and loaded from the open-source Hugging Face `transformers` library [15]. In addition to Pratihast's findings, the SUPERB benchmark, which we use for our main evaluation work, also found these models to be among the strongest models. They are also highly comparable because their Base+ architecture and size are the same, giving us a controlled study. We used the Base+ versions so that extraction would be faster to run on a laptop.

Each clip was loaded at 16 kHz and passed through each model in evaluation mode with no gradient updates, and the same pretrained model weights were used with no weight updates, as we are not focusing on training. This was also done in SUPERB [8] and Pratihast's study [10]. All 13 hidden states were kept: the CNN encoder output (layer 0) and the outputs of the 12 Transformer layers. Each layer's frame sequence was mean-pooled over time to give one 768-dimensional vector per layer, producing an array of $1440 \times 13 \times 768$ per model, which was saved once and reused.

Mean pooling was chosen for simplicity and comparability, even though [10] shows it can lose timing detail. Unlike the learned weighted sum of SUPERB [8] and Pepino et al. [13], each layer is kept separate so that it can be probed on its own (Section 3.4), as one of the aims of this project is to identify the best layers.

3.4 Classifier and evaluation protocol

This is the machine learning portion of our project, and every data processing tool and model was imported from the scikit-learn library [16]. Our chosen model is multinomial logistic regression, because we wanted to focus on the acoustic and SSL features it uses for emotion prediction instead of trying to find the best machine learning model.

The pipeline below describes how we progressed from data standardisation (to avoid penalising small-scale features too much) all the way to using test data to evaluate the classifier trained on each set of features (two feature sets from the SSL models and one hand-crafted feature set), in `run_classifiers.py`:

Standardisation (StandardScaler) was followed by multinomial logistic regression with balanced class weights (to account for neutral having half as many clips) and a fixed regularisation strength $C = 0.01$. A linear probe was chosen deliberately. It measures how accessible the emotion information is in each representation, rather than how much a powerful classifier can extract, which is the standard way of comparing frozen

representations [8], [11]. C was fixed for all 27 systems and not tuned on test data; strong regularisation suits 768-dimensional features with only about 1,380 training clips per fold, and it also kept the 648 model fits (27 systems $\times$ 24 folds) practical in run time. The scaler and classifier were fitted on training data only within each fold.

Evaluation was then performed with leave-one-actor-out cross-validation. This ensured that the test set had actors who were not present in the training set. This directly addresses the problem of speaker leakage mentioned in Section 2.5, ensuring our predictions are not dependent on individual identity, but rather purely acoustic and voice based. The predicted label and the human agreement scores were all kept, which will form the ground truth for our classifier predictions.

Macro-F1 was our main simple metric for direct visual comparison, and secondly we used EMO-SUPERB [9] because it weights every emotion equally; accuracy is also reported. Every actor was given a macro-F1 score, and the mean of all 24 actors' (audio clips') scores was computed with a 96 percent confidence interval. It should be clarified that when we refer to actors here, we are referring to each audio clip in the dataset the model is tested on; we are not rating human listeners here. Layer comparisons were made between the best middle layer, layer 0 and layer 12 for features from WavLM and UniSpeech-SAT. In this layer comparison, the Wilcoxon signed-rank test was used to make an actor-by-actor comparison instead of just averaging across all 24 actors, because if actors vary a lot, averaging across them would make us think these differences are caused by differences between the layers, not the actors. Finally, we compared the SSL performance against our hand-crafted baseline feature set.

3.5 Human–machine comparison

Next, we compared humans against our model performance. As mentioned in the dataset section, for a particular audio clip (actor), 10 humans listened to it and each emotion was given a score based on what proportion of the listeners perceived it as the correct emotion. Naturally, for a clip, the emotions with the lowest scores highlight where humans struggle the most. The particular clips where humans struggled with the emotion being conveyed are expected to have distinctive patterns in their acoustic features and spectrograms. We wanted to see to what extent models trained on SSL features struggle with clips with the same features and spectrograms. Hence, we examine how well human ambiguity in emotion detection matches SSL-based models.

Model performance was first given a model confidence score (p_{true}). This tells us how sure the classifier was that a particular emotion was correct for a particular audio clip, and ranges from 0 to 1, similar to human agreement scores. For the acoustic baseline and every layer of both models, alignment was measured, which tells us how closely our models match human ambiguity. Spearman rank correlation ρ was used to measure this alignment, between p_{true} from the model and agreement scores from human listeners across all 1,440 clips. This correlation test is appropriate here because it only compares the ranking of clips, so it is not affected by large outlier scores, and it does not require a normal distribution of scores. 95% confidence intervals were estimated from 500 bootstrap resamples of clips.

However, purely observing a high ρ, for example, only tells us whether the models struggle with the same emotion as humans, without giving clip-specific agreement, as clips from each actor will have specific acoustic features and spectrograms, which we want to investigate if we are to relate our study to vocal measurements instead of only emotion-wise comparison. To overcome this, ρ was also computed within each emotion and averaged, and the per-emotion mean human agreement was correlated with the per-emotion mean model confidence.

To look at agreement and disagreement more directly, clips were sorted into quadrants to see how many clips humans struggled with the models got right, how many they got wrong, and vice versa. A clip was human-easy if agreement was at least 0.8 and human-hard if agreement was at most 0.4, and model-right or model-wrong depending on the predicted label. Confusion matrices were then used to compare which emotions the models mixed up. To test whether model errors stay within similar arousal levels, emotions were grouped using the circumplex model [17] into low (neutral, calm, sad), mid (disgust) and high (happy, surprised, angry, fearful) arousal. The share of errors that stayed within the same group was compared with

the 32% expected by chance.

The effect of intensity was tested by pairing each strong clip with its normal-intensity version from the same actor, emotion, statement and repetition (672 pairs). The changes in human agreement, model confidence, F0 and loudness were compared with paired Wilcoxon tests. As a robustness check, alignment was also recomputed with calm and neutral clips removed, because their lenient human scoring (Section 3.1) could distort ρ.

Finally, clips where humans and models disagreed were examined as case studies in MATLAB. For each one, the waveform, spectrogram, F0 contour and intensity contour were plotted to see whether the acoustic features explain the difference.

3.6 Tools learned in this study

New tools learned for this project include the Hugging Face `transformers` library for loading and running pretrained speech models, MATLAB Audio Toolbox functions (`pitch()`, `harmonicRatio()`, `mfcc()`), scikit-learn for cross-validated classification, and SciPy for Spearman correlation and Wilcoxon tests. All code, fold assignments and per-clip predictions are in the GitHub repository, so every figure can be regenerated from the saved results.

4 Results

All results are speaker-independent: every prediction comes from the fold in which that clip's actor was held out of training. Chance level for eight balanced classes is a macro-F1 of 0.125.

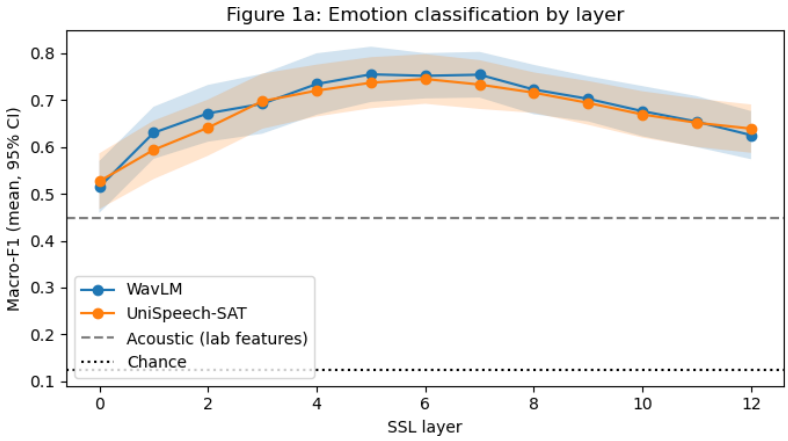
4.1 Emotion classification across layers

Both SSL models followed the same pattern across depth (Figure 1a). Macro-F1 rose from about 0.52 at layer 0 in both models to a peak in the middle of the network, 0.755 at layer 5 for WavLM and 0.745 at layer 6 for UniSpeech-SAT, and then fell to 0.63–0.64 at layer 12. The acoustic baseline reached 0.447, well above chance but far below every SSL layer.

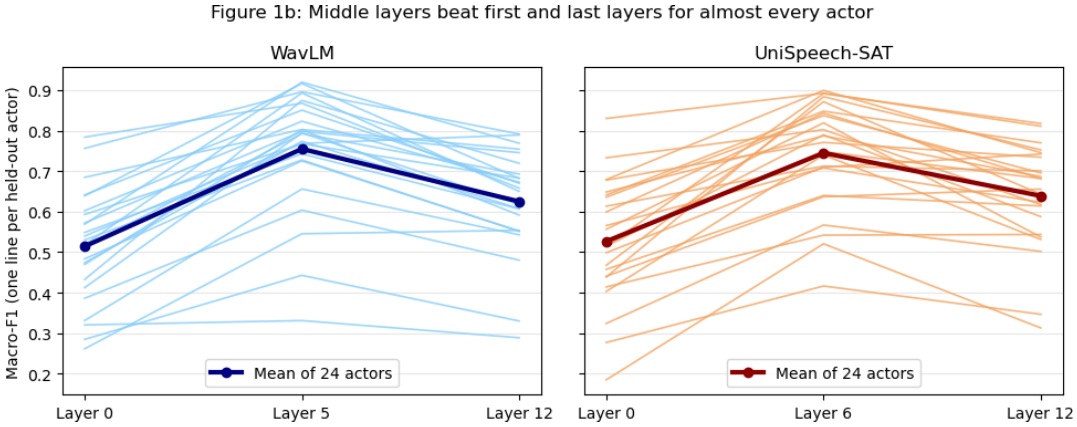
The per-actor view (Figure 1b) shows that this pattern is consistent across speakers rather than driven by a few actors. The best middle layer beat layer 0 for all 24 actors in both models (mean gain +0.24 for WavLM and +0.22 for UniSpeech-SAT), and beat layer 12 for 22 of 24 actors (WavLM, +0.13) and 21 of 24 actors (UniSpeech-SAT, +0.11) (paired Wilcoxon, $p < 0.001$; Table 3). WavLM layer 5 beat the acoustic baseline for all 24 actors by an average of 0.31. The two models did not differ significantly from each other: WavLM was ahead for 16 of 24 actors by an average of only 0.01 ($p = 0.49$). Performance varied widely between actors: at WavLM layer 5, per-actor macro-F1 ranged from about 0.44 to 0.92, so a single average hides considerable speaker-to-speaker variation.

Table 3: Paired comparison of per-actor macro-F1 across the 24 held-out actors.

Comparison	Mean ΔF1	Actors better	Wilcoxon p
WavLM L5 vs WavLM L12	+0.130	22/24	< 0.001
WavLM L5 vs WavLM L0	+0.240	24/24	< 0.001
UniSpeech-SAT L6 vs UniSpeech-SAT L12	+0.106	21/24	< 0.001
UniSpeech-SAT L6 vs UniSpeech-SAT L0	+0.218	24/24	< 0.001
WavLM L5 vs UniSpeech-SAT L6	+0.010	16/24	0.491
WavLM L5 vs acoustic baseline	+0.307	24/24	< 0.001



(a)



(b)

Figure 1: (a) Speaker-independent macro-F1 by layer for WavLM and UniSpeech-SAT; shaded bands show the 95% confidence interval of the mean across the 24 held-out actors. Dashed: acoustic baseline (a single feature set, not layer-dependent); dotted: chance. (b) Macro-F1 of each actor at layer 0, the best layer and layer 12.

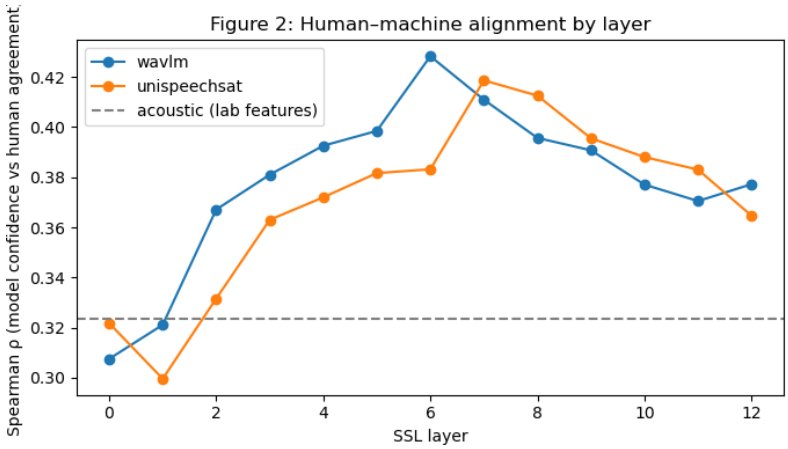


Figure 2: Spearman ρ between model confidence in the intended emotion and listener agreement, by layer.

4.2 Alignment with human perceptual agreement

Model confidence was positively correlated with listener agreement for every system (Figure 2). Alignment followed the same rise-and-fall shape as classification, peaking at $\rho = 0.428$ [0.382, 0.469] at layer 6 for WavLM and $\rho = 0.419$ [0.374, 0.462] at layer 7 for UniSpeech-SAT, compared with $\rho = 0.324$ [0.277, 0.366] for the acoustic baseline. Layer 0 of both models aligned no better than the acoustic baseline ($\rho = 0.307$ and 0.322). The alignment peak sits one layer deeper than the classification peak in both models, although the

confidence intervals of neighbouring layers overlap.

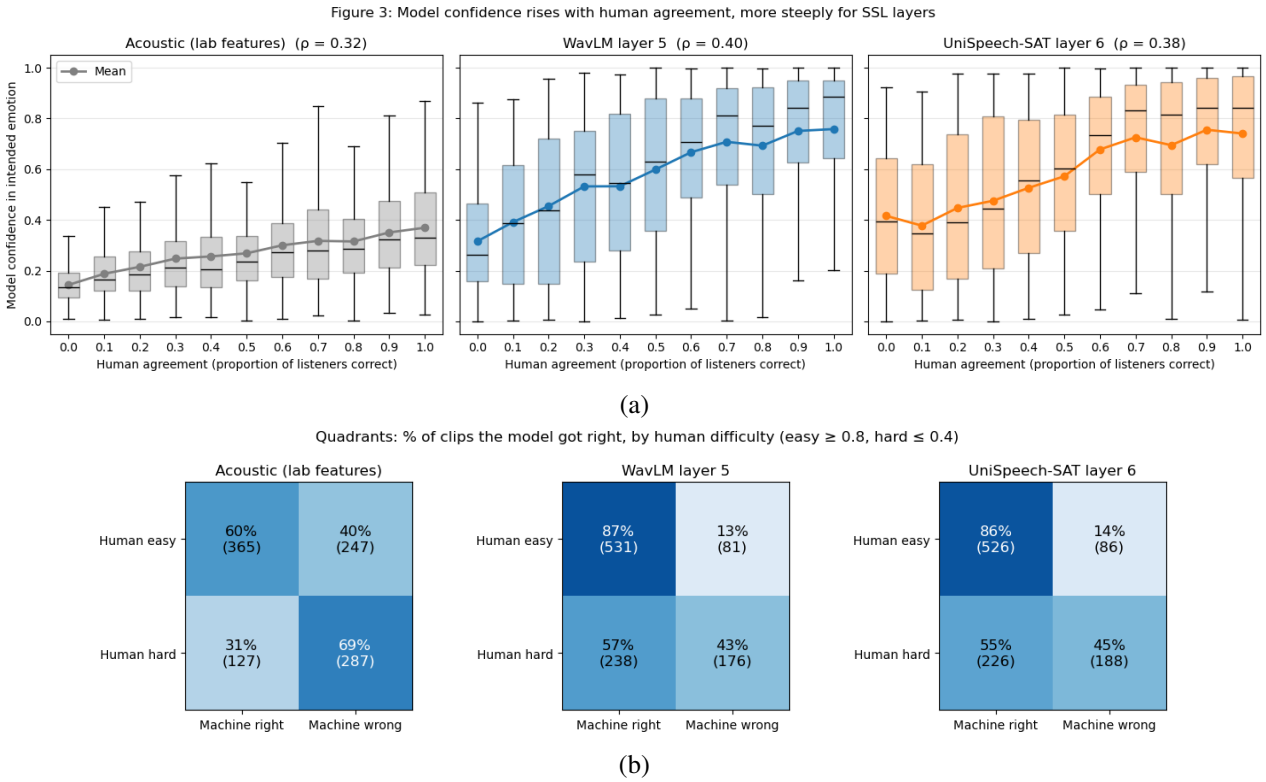


Figure 3: (a) Model confidence in the intended emotion for clips grouped by listener agreement (boxes: median and interquartile range; line: mean), for the acoustic baseline, WavLM layer 5 and UniSpeech-SAT layer 6. (b) Percentage of human-easy (agreement ≥ 0.8) and human-hard (agreement ≤ 0.4) clips each model classified correctly, with clip counts.

Figure 3 shows the same relationship clip by clip. As listener agreement increases, median model confidence increases for all three systems, but the spread within each agreement level is large, which is consistent with a moderate rather than strong correlation. The SSL models are more confident than the acoustic baseline at every agreement level.

We performed a quadrant analysis (Figure 3b), which gives the same picture in terms of correct and incorrect predictions. Of the 612 clips listeners found easy (agreement ≥ 0.8), WavLM layer 5 classified 87% correctly and UniSpeech-SAT layer 6 86%, against 60% for the acoustic baseline. Of the 414 clips listeners found hard (agreement ≤ 0.4), WavLM still classified 57% correctly and UniSpeech-SAT 55%, against 31% for the acoustic baseline, so it agrees moderately well with how humans struggled with emotions. All systems were less accurate on human-hard clips, but the SSL models were correct on more than half of them.

Figure 2 shows overall alignment at every layer; Table 4, however, takes the best-aligned layer of each model and separates that alignment into clip-level and emotion-level components. For example, in the mean within-emotion column of Table 4, ρ values are computed from each set of audio files that share the same emotion. These ρ values are then averaged.

The per-emotion column was derived as follows: a score is given to each emotion for humans, based on how much agreement humans had for that emotion. Then a score is given to each emotion for the model, based on how accurately that emotion is predicted. These two columns are sorted by emotion and placed side by side. If the emotion ranking (based on these scores) of human agreement matches the order of the model scores, the per-emotion alignment gets a high score. This is how the per-emotion column is devised, to observe how well models excel and struggle with the same emotions as humans.

The results of this separation are shown in Table 4. At the emotion level, the acoustic baseline matched listeners closely ($\rho = 0.83$), ranking the eight emotions in almost the same order of difficulty, while WavLM (0.40) and UniSpeech-SAT (0.48) matched less well. At the clip level the pattern reversed: the mean within-

emotion ρ was 0.40 for WavLM and 0.39 for UniSpeech-SAT, but only 0.22 for the acoustic baseline. The hand-crafted features therefore reflect which emotions listeners find difficult, while the SSL features better reflect which individual clips within an emotion listeners find difficult.

Table 4: Human–machine alignment at the clip and emotion levels, at each model's best-aligned layer. *Per-emotion ρ uses the best-classifying layers (WavLM 5, UniSpeech-SAT 6).

System	Overall ρ (1,440 clips)	Mean within-emotion ρ	Per-emotion ρ (8 emotions)*	ρ without calm/neutral
Acoustic baseline	0.324	0.216	0.83	0.388
WavLM (layer 6)	0.428	0.401	0.40	0.515
UniSpeech-SAT (layer 7)	0.419	0.385	0.48	0.529

Removing calm and neutral clips, whose human scores are lenient (Section 3.1), raised alignment for all systems to 0.515 (WavLM), 0.529 (UniSpeech-SAT) and 0.388 (acoustic). The lenient scoring therefore lowers the measured alignment rather than inflating it.

4.3 Confusion patterns

Errors were concentrated among emotions of similar arousal (Figure 4a). For the acoustic baseline, sad was the weakest class (22% correct) and was confused equally with neutral and calm (21% each), while happy was correct in only 32% of cases. The SSL models were stronger on every class, reaching 0.71–0.85 on most emotions. Their weakest class was also sad (0.57 for WavLM and 0.59 for UniSpeech-SAT), most often mistaken for calm (18% and 13%), and WavLM also labelled 15% of neutral clips as calm. Their remaining errors followed the same structure: 57% of WavLM and 57% of UniSpeech-SAT errors stayed within the same arousal group, compared with 51% for the acoustic baseline and 32% expected by chance (Figure 4c).

Listeners and models found partly different emotions hardest (Figure 4b). Listener agreement was highest for neutral (0.91) and calm (0.77), partly because of the lenient scoring, and lowest for happy (0.37), in line with the low audio-only happiness accuracy reported in [14]. The SSL models did not share the difficulty with happy, which they recognised in 71% of cases, but sad was hard for listeners (0.48) and models (0.57–0.59) alike. The SSL models therefore match listeners on some emotions but not others, which is consistent with their moderate emotion-level alignment in Table 4.

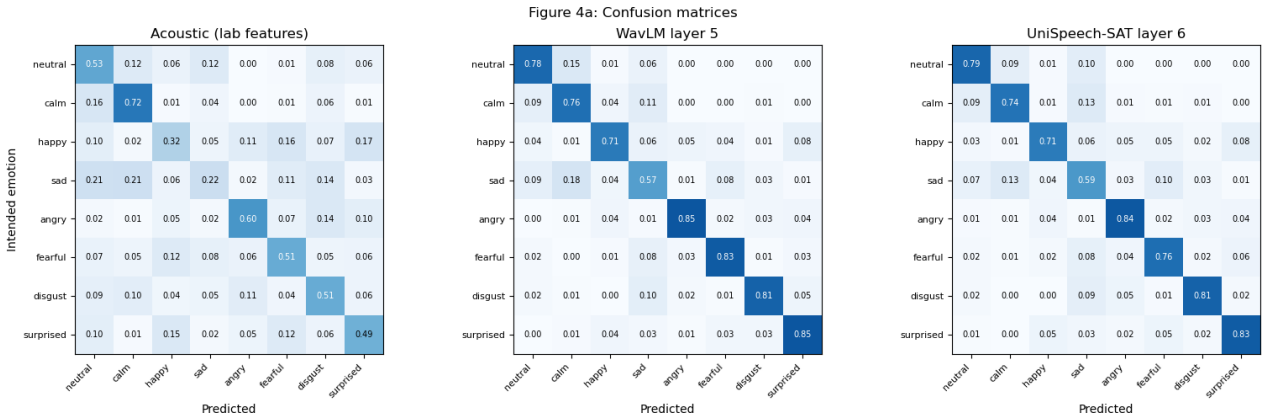
4.4 Effect of expression intensity

Strong portrayals were acoustically more extreme than their normal-intensity pairs: mean F0 was 52 Hz higher (199 Hz to 250 Hz, about 4 semitones) and mean intensity 7.2 dB higher (Figure 5). Listeners recognised strong portrayals more easily, with agreement rising from 0.53 to 0.67 (+0.14). Both SSL models showed a similar gain in confidence (+0.13 for WavLM and +0.12 for UniSpeech-SAT), while the acoustic baseline gained less (+0.09). All changes were significant (paired Wilcoxon, $p < 0.001$). Humans and models therefore benefited almost equally from the stronger acoustic cues.

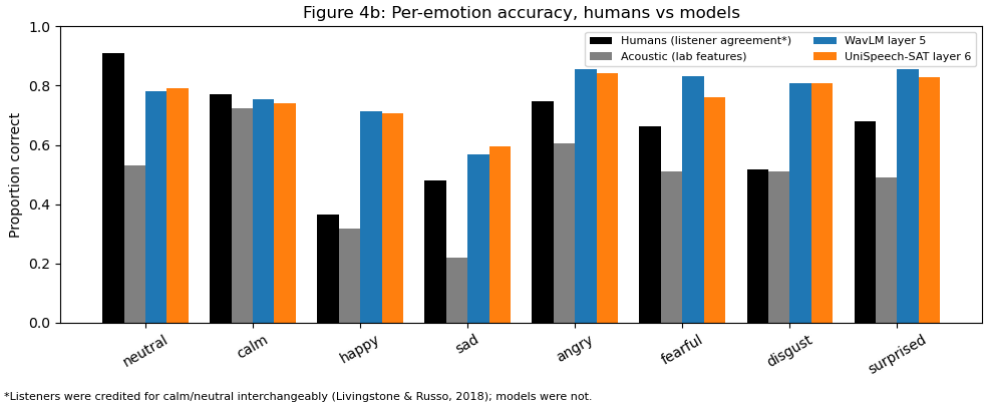
In other words, when the emotion is acted more strongly, the cues get bigger, and humans and SSL models get better by almost exactly the same amount. This is another way in which the SSL models behave like human listeners: both respond to cue strength in the same way. The acoustic baseline benefits less, even though it directly measures pitch and loudness.

4.5 Acoustic case studies

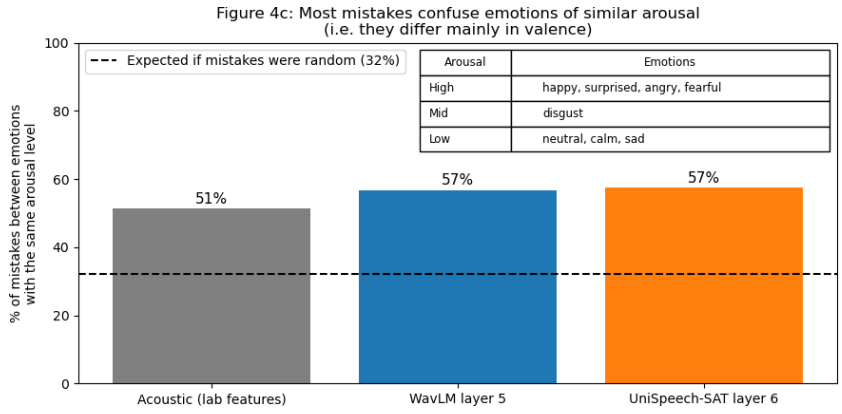
This section examines clips where both SSL models agreed with each other but not with listeners, and uses interpretable acoustic measures to explain these disagreements. Thirteen clips were recognised by at least 90% of listeners yet given a probability below 0.2 by both models, and almost all of them were calm portrayals by actor 13, which the models labelled as neutral or sad. In the opposite direction, 31 clips were recognised by at



(a)



(b)



(c)

Figure 4: (a) Row-normalised confusion matrices for the acoustic baseline and the best layer of each SSL model. (b) Per-emotion accuracy of listeners and models (listener scores for calm and neutral are lenient). (c) Share of errors that stay within the same arousal group, against the 32% expected by chance.

most 30% of listeners but given a probability above 0.8 by both models; these were mostly happy and disgust portrayals, two of the emotions listeners found hardest. Figure 6 examines one example of each.

Actor 13's calm clip was recognised by every listener, yet both SSL models labelled it neutral. Its pitch and loudness are very similar to the same actor's neutral clip (mean F0 135 Hz vs 122 Hz; mean intensity -55.2 dB vs -54.1 dB), and both show a similar falling F0 contour. Because listeners who answered neutral were also scored as correct, this clip may reflect a confusion that humans and models share but that the human scoring hides.

In the second case, an actor 7 happy clip was recognised by only 2 of 10 listeners, while both models classified it correctly with high confidence (0.81 and 0.85). Compared with actor 13's clips, it is higher in pitch (mean

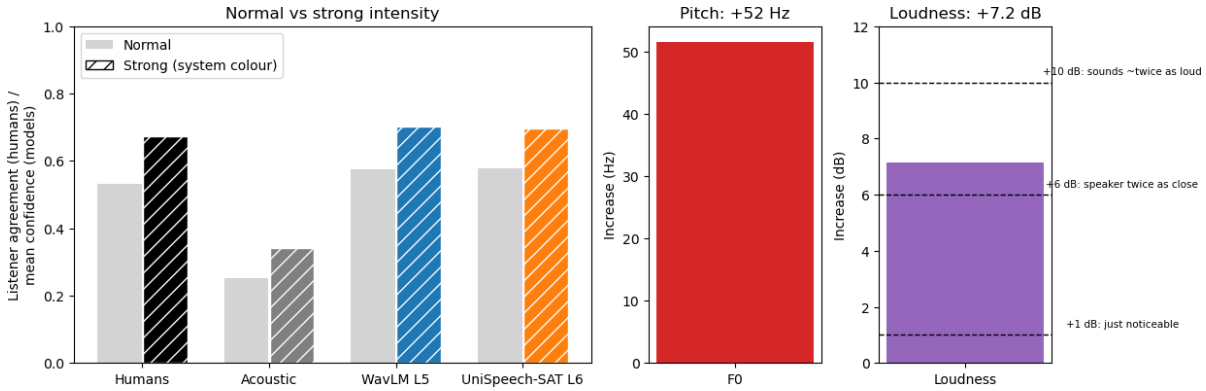


Figure 5: Normal versus strong intensity for 672 matched clip pairs: listener agreement, model confidence (p_{true}), and the accompanying change in F0 and loudness.

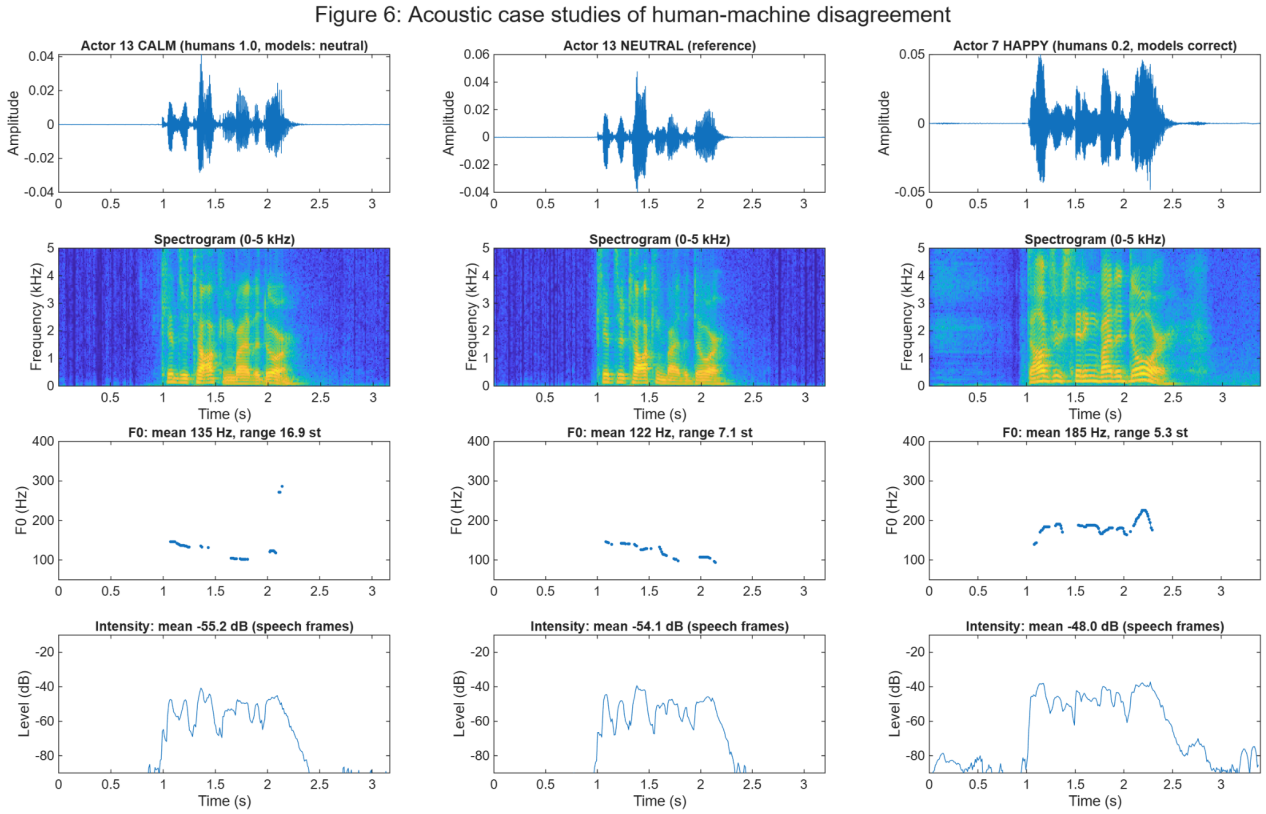


Figure 6: Waveform, spectrogram (0–5 kHz), F0 contour and intensity for actor 13 calm (listeners 1.0*, both models predicted neutral), actor 13 neutral (reference), and actor 7 happy (listeners 0.2, both models correct). *Calm and neutral responses were both scored as correct.

F0 185 Hz) and louder (−48.0 dB), and its spectrogram shows strong harmonics across the whole utterance. An identical copy of the same recording was rated 0.7 by a different group of listeners, which shows how much per-clip agreement can vary with only about ten raters.

4.6 Sources of error

Several factors limit these results. Human agreement is based on about ten raters per clip, so it is noisy, and this caps the correlation any model can reach. The lenient calm and neutral scoring makes human and model accuracy not directly comparable for those two classes. Each actor contributes only 60 test clips, so per-actor macro-F1 varies considerably. Mean pooling removes timing information that may help with some emotions. Finally, all recordings are portrayals of two fixed sentences, so the results may not carry over to spontaneous speech.

5 Discussion

In this section we relate our results to the theories discussed in the literature review and interpret them.

5.1 Where is emotion information in SSL models?

Emotion information in both models was most accessible in the middle of the network. Macro-F1 peaked at layer 5 in WavLM and layer 6 in UniSpeech-SAT. The middle layer beat the first layer for every held-out actor and the last layer for almost every actor. This agrees with the layer-wise literature. Pasad et al. [11] showed that the upper layers of a pretrained SSL model specialise toward the pretraining target, and Pepino et al. [13] found that learned layer weights for emotion favoured the middle layers of wav2vec 2.0. The present results extend these findings to WavLM and UniSpeech-SAT. They confirm them with a direct per-layer probe, in which every layer is evaluated with the same classifier and compared actor by actor, rather than inferring layer importance from learned weights, which EMO-SUPERB [9] found hard to interpret.

Performance fell from the best middle layer to the final layer by 0.11 to 0.13 macro-F1, a large enough difference to affect how SSL models should be used. Pratihast [10], the closest prior work to this study, extracted features only from the final layer of each SSL model. Studies that use only the final layer may therefore underestimate how much emotion information these models contain, although this cannot be confirmed for [10] directly, since it used different datasets and mostly larger models. At the other end of the network, even layer 0, the CNN output before any Transformer context, outperformed the 36 hand-crafted features (0.52 against 0.45). A plausible explanation is that layer 0 acts as a detailed learned spectrogram, describing spectral shape and voice quality in 768 dimensions where the hand-crafted set keeps only summary statistics. The following Transformer layers then add context across the utterance, which lifts performance to about 0.75. This explanation was not tested directly.

The two models performed almost equally ($p = 0.49$). This matches the previous studies' IEMOCAP results [5], [7]. Hence we were able to validate their conclusion that SSL models that contain speaker identity information (UniSpeech-SAT) will not perform better than models that have voice-specific features (WavLM), so we do not have to know the identity of a speaker for these predictions, and it is sufficient to know the features that shape a person's voice, allowing our study to be applied to anonymous test subjects just by examining the distal features of their voice. We also ensured that actor identity did not leak into the test set for either model by using leave-one-actor-out evaluation.

5.2 Do SSL representations reflect human perceptual ambiguity?

Model confidence followed listener agreement moderately. The strongest alignment was at WavLM layer 6 ($\rho = 0.43$) and UniSpeech-SAT layer 7 ($\rho = 0.42$), compared with 0.32 for the acoustic baseline. In the previous section we observed that the middle SSL layers performed best for classification, and interestingly, the middle layers also performed best for alignment of models with human ambiguity. The alignment peak was one layer deeper than the classification peak in both models, but the confidence intervals of neighbouring layers overlap, so this difference should not be over-interpreted.

A model–human alignment correlation of 0.43 is much stronger when we consider the subjective nature of human interpretation, especially of emotion, compared with perfect mathematical models, which would interpret this value as much weaker. The human agreement column in the RAVDESS dataset had an average score of 0.62. In addition, the RAVDESS study reported Fleiss' kappa of 0.53 to 0.62, which measures how often human listeners agreed in their answers, whether correct or not. So humans themselves cannot score even close to a ρ of 1 when trying to align with the predictions of another person. So if our models were aiming to be as good as humans, the true goal value of ρ is well below 1. Furthermore, there was an observed case of an identical clip appearing twice in the dataset, where one copy had a human agreement score of 0.7 while the other had 0.2, showing further how individual humans differ in their perception of emotion.

Another informative result is how differently the systems behave at the clip level and the emotion level. Clip-level alignment was measured by correlating model confidence with human agreement across the clips of each emotion separately and averaging the eight results, so it shows whether a model finds the same

individual recordings hard as listeners do. Emotion-level alignment was measured by averaging human agreement and model accuracy for each emotion and correlating these eight pairs, so it shows whether a model finds the same emotion categories hard. The hand-crafted features matched listeners on which emotions are hard ($\rho = 0.83$) but not on which individual clips are hard ($\rho = 0.22$). The SSL features showed weaker emotion-level agreement (0.40 to 0.48) but nearly twice the clip-level agreement (0.39 to 0.40) compared with the hand-crafted acoustic features. This suggests that, compared with the hand-crafted features, the SSL representations capture more of the clip-specific detail that makes a particular portrayal clear or ambiguous to listeners, rather than only the broad difficulty of each emotion category. The SSL models therefore align with listeners moderately at both levels, while the hand-crafted features align strongly at the emotion level but weakly at the clip level.

One explanation for this could be that hand-crafted acoustic features are mostly based on average measurements, including mean pitch, mean loudness, duration and average MFCCs, and these summaries are good at capturing big differences between emotions (such as anger being loud and high) and hence match well with listeners at the emotion level. But within emotions, averages are very similar, and other ambiguous, finer features are required (such as breathy or tense voice quality), allowing SSL models, with their rich, extensive features (768 numbers per layer) that do not throw away much detail, to perform better at the clip level, even between clips with the same emotion.

The SSL models classified 55 to 57% of human-hard clips correctly and recognised happy 71% of the time, whereas human agreement on happy (an emotion humans find one of the hardest) was only 37%. This, however, should not lead us to assume that our SSL models outperformed humans at all. Human perception is trained on everyday sounds and conversations with vast variations in tone and context, while the models in this study were trained on the niche RAVDESS dataset with 23 actors speaking the same two sentences. If the test set was obtained from an entirely different dataset or context, it may have largely outperformed human perception.

Where the models did fail, their errors followed the arousal structure described in the perception literature: between 51% (acoustic) and 57% (SSL) of errors stayed within the same arousal group, against 32% expected by chance. The most common SSL confusion, sad heard as calm, also mirrors one of the most frequent listener confusions reported in [14], sad heard as neutral or calm. This is consistent with the vocal emotion literature, where arousal is well signalled by pitch, loudness and rate, while valence depends on subtler voice-quality cues and is harder to recognise [1], [3].

5.3 What interpretable acoustics reveal

The acoustic baseline was far less accurate than either SSL model, but its errors are easier to explain. Its strong emotion-level alignment with listeners suggests that the features it measures, mainly pitch, loudness, timing and coarse spectral shape, are the cues that make some emotion categories easier for listeners than others, as the lens model and cue profiles in [1] and [2] predict. Its weak clip-level alignment suggests that utterance averages of these cues miss the voice-quality and temporal details that make an individual portrayal ambiguous. This matches Pepino et al. [13], where adding eGeMAPS features still improved an SSL system, and suggests that the two kinds of representation are complementary rather than one replacing the other.

The intensity analysis links the two views. Strong portrayals were about 4 semitones higher and 7.2 dB louder than their normal pairs, in line with Major and Chatterjee's acoustic analysis of RAVDESS [4]. Listeners gained 0.14 in agreement and the SSL models gained 0.12 to 0.13 in confidence, while the acoustic baseline gained only 0.09. Humans and SSL models therefore benefited from stronger acoustic cues by almost the same amount, which suggests that both respond to the same increase in cue strength. The acoustic baseline gained less even though it measures pitch and loudness directly, which suggests its limitation lies less in missing these cues than in separating emotions from 36 utterance-level averages.

The case studies show why some disagreements are not real disagreements. All listeners were credited with recognising actor 13's calm clip, yet its pitch, loudness and F0 contour are very close to the same actor's neutral clip. Because listeners who answered neutral were also counted as correct, the models' neutral prediction may reflect the same confusion listeners made, hidden by the scoring rule. In the opposite case, the

actor 7 happy clip that only two listeners recognised has clearly raised pitch and loudness, and an identical copy received 0.7 agreement, so its low score reflects rater noise as much as genuine ambiguity. Both cases show that per-clip human agreement must be interpreted with its scoring rules and sample size in mind.

5.4 Limitations

There are several limitations that may restrict how far our conclusions generalise. One of the main ones is training and evaluating on the RAVDESS dataset, which has much less variation than everyday conversation. Two fixed sentences were spoken, and all speakers were 24 young American actors who exaggerate phrases much more. Whether our conclusions transfer to more realistic datasets of everyday speech may be checked in further study. With 60 clips per actor, per-actor macro-F1 is noisy, which is why paired tests across actors were used. Mean pooling also discards timing information that Pratihast [10] showed can help, especially for valence.

Another limitation was that the data only contained information on what proportion of human listeners picked the correct emotion, but did not contain data on which emotions listeners chose when they were incorrect. It was therefore not possible to do a clip-by-clip analysis of where the humans and models failed alike. Listeners also got credit for answering with unlisted emotions, through manual corrections, but our models did not have this benefit. On the acoustic side, automatic pitch tracking still produces occasional octave errors despite voicing checks and median filtering, as seen in the calm clip of Figure 6, which adds noise to the F0 features. And while one of our other limitations was using only one classifier model, logistic regression with a fixed learning rate, this was essential to have a controlled experiment that focuses on feature effectiveness in prediction and alignment with human ambiguity.

5.5 Future work

Each extension below follows from a specific finding or limitation of this study.

First, this study tested each layer on its own and found that the middle layers work best. A natural next step is to combine several middle layers, either by joining their features or by letting the model learn how much weight to give each layer, and to check whether this improves both accuracy and alignment with listeners.

Second, mean pooling averages each clip into a single vector, which loses information about how the voice changes over time, such as a rising pitch at the end of a sentence. Pooling methods that keep some of this timing, such as attention-based or two-step pooling [10], could help with emotions that are hard to separate, such as sad and calm.

Third, the models here were trained only on the intended emotion, even though listeners often disagreed. Training on soft labels, which give each clip the proportion of listeners who chose each emotion, as in EMO-SUPERB [9], would test whether models that learn human uncertainty directly match listener ambiguity more closely.

Fourth, the hand-crafted features matched listeners on which emotions are hard, while the SSL features better matched which individual clips are hard (Section 5.3). Combining the two feature sets would test whether they complement each other.

Finally, RAVDESS contains acted speech of two fixed sentences, rated by about ten listeners per clip. Repeating the analysis on natural conversational speech with more raters per clip would give more reliable human scores and show whether these findings hold in real-world speech.

6 Conclusion

This study examined whether frozen WavLM and UniSpeech-SAT representations reflect human perceptual agreement on emotional speech, which layers hold the most emotion information, and how they compare with interpretable hand-crafted acoustic features. Each layer of both models was probed with the same linear classifier on the RAVDESS audio-only speech set, evaluated speaker-independently with leave-one-actor-out cross-validation, and compared clip by clip with listener agreement.

For RQ2, emotion information was most accessible in the middle layers of both models, with macro-F1 peaking at about 0.75 at layers 5 and 6. This confirmed the middle-layer advantage suggested by earlier layer-wise studies, and showed that using only the final layer can understate these models by 0.11 to 0.13. The middle layers also aligned best with listeners. For RQ1, model confidence followed listener agreement moderately ($\rho \approx 0.43$), a meaningful result given the noise in ratings from about ten listeners per clip, and higher still once the leniently scored calm and neutral clips were removed. When the models were wrong, their errors mostly stayed between emotions of similar arousal, as in human perception. For RQ3, the SSL features clearly outperformed the 36 hand-crafted features (0.75 against 0.45 macro-F1) and showed stronger overall alignment. The two representations also related to listeners in different ways: the hand-crafted features reflected which emotions listeners find difficult, while the SSL features better reflected which individual recordings listeners find ambiguous.

The main contribution of this study is using per-clip human agreement as a reference for testing models, rather than only as a training target, and showing at which depth of the network this alignment is strongest. The conclusions are limited to acted speech of two fixed sentences rated by about ten listeners per clip. Future work should combine middle layers, keep timing information instead of mean pooling, train on listener response distributions, and test whether the clip-level alignment found here holds in natural conversational speech.

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- Debugging and refining code, including data preprocessing, the MATLAB feature extraction (based on course lab examples) and plotting code.
- Formatting and structuring the report, and improving tone and wording to match scientific language and paper style.
- Explaining concepts from the literature articles.
- Helping interpret plots and statistical results and understand them in terms of theory.